PALM-mGWAS: end-to-end microbiome genome-wide association analysis and meta-analysis

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Abstract

Summary: Microbiome genome-wide association studies require software that can generate cohort-level summary statistics, combine results across studies, and return interpretable outputs without extensive custom scripting. We present PALM-mGWAS, an R package and command-line workflow for microbiome GWAS based on the PALM framework. PALM-mGWAS performs input checking, taxon-level and sequencing-depth summarization, PALM-based null model fitting, per-study genome-wide summary statistic generation, fixed-effect meta-analysis across studies, and downstream visualization. The workflow accepts PLINK and VCF genotype inputs, writes standardized file-per-taxon results, and is designed for both local and high-performance computing environments.

Availability and implementation: PALM-mGWAS is implemented in R and shell workflow wrappers. Source code, example data, and documentation are available in the project repository.

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1 Introduction

Microbiome genome-wide association studies analyze host genetic effects on microbial abundance traits. In multi-cohort settings, association testing is only one part of the problem. Practical analyses also require input validation, cohort-level standardization of summary statistics, cross-study aggregation and result visualization. Existing application notes have typically emphasized either microbiome-focused single-study analysis or efficient meta-analysis of genetic score statistics. PALM-mGWAS was developed to connect these steps in one workflow.

PALM-mGWAS implements a PALM-based mGWAS pipeline with five stages: auxiliary summary generation, null model fitting, per-study score-based association testing, fixed-effect meta-analysis and visualization. The package is designed to produce standardized outputs that can be reused across cohorts and downstream reports.

2 Methods

3 Results

3.1 Implementation

PALM-mGWAS can be run locally with a `pixi`-managed environment or through containerized execution on HPC systems. The workflow is distributed as modular shell wrappers for direct execution or scheduler submission. Each stage produces explicit on-disk outputs, allowing per-step inspection and profiling.

3.2 Performance

Step2 performance was evaluated on three cohorts using 8 CPUs per task, 64 GB requested memory, and `minMAF=0.1`. Association testing was parallelized by chromosome as a 22-task Slurm array, with one task assigned to each autosome. Runtime and peak memory were recorded for all 22 tasks in each cohort. Cohort sizes, taxon counts, variant counts, and the resulting runtime and memory summaries are shown in Table 1.

3.3 Visualization outputs

The visualization module summarizes meta-analysis results in forms suitable for downstream reporting. For a selected taxon, PALM-mGWAS generates a Manhattan plot together with the corresponding QQ plot. For a selected SNP, the workflow renders one forest plot per retained taxon, with each plot reporting cohort-specific effect estimates together with the combined meta-analysis estimate. At the global level,

Table 1: Observed `step2` performance of PALM-mGWAS across three cohorts. Runtime and memory are summarized across 22 completed array tasks per cohort.

Dataset	Samples	Taxa	Variants	Median elapsed	Max elapsed	Median MaxRSS	Max MaxRSS
INSPIRE	503	139	6 174 920	01:24:23	03:15:40	33.33 GB	38.54 GB
URECA	99	188	7 896 716	00:41:12	01:43:46	9.82 GB	11.29 GB
WISC	88	139	5 758 326	00:25:17	01:00:56	6.50 GB	7.35 GB

the package also produces a combined significant-hit summary that reports the top associated taxon for each lead SNP passing the requested threshold.

3.4 Comparison with existing software

mbQTL provides a practical point of comparison because it is designed for microbial quantitative trait locus estimation, that is, host variant–taxon association analysis within individual cohorts. A further difference is workflow structure. *mbQTL* operates more directly from the original abundance, covariate, and genotype inputs, whereas PALM-mGWAS is organized as a staged mGWAS pipeline in which downstream analyses consume intermediate summary-statistic tables produced by earlier steps. In particular, visualization in PALM-mGWAS is driven by cohort-level and meta-analysis summary-statistic outputs rather than by the raw input tables. Under this framing, *mbQTL* ran faster on datasets of the same scale, which is consistent with the fact that its core computation is simpler. *mbQTL* focuses on within-cohort association estimation, whereas PALM-mGWAS adds the overhead of PALM-based summary generation together with downstream standardization, meta-analysis, and visualization.

4 Discussion

PALM-mGWAS provides a compact implementation for microbiome GWAS that links cohort-level PALM analysis to cross-study meta-analysis and visualization. The main strength of the package is workflow integration: the same framework handles preprocessing, null model fitting, genome-wide summary-statistic generation, study aggregation and reporting. This makes the software practical for consortium-scale analyses where standardized outputs are required across cohorts.

Because the package combines stages that are often handled separately, total runtime should be interpreted as the sum of preprocessing, cohort analysis and downstream synthesis rather than as the cost of association testing alone. The observed `step2` results show that runtime and memory scale sharply with cohort size and variant count: the largest cohort examined here, INSPIRE, required roughly three times the median elapsed time and more than five times the median memory of WISC. Relative to *mbQTL*, PALM-mGWAS required more memory in the cohort-level testing stage, but completed successfully across all three cohorts and produced directly meta-analyzable outputs. Under this framing, PALM-mGWAS can be evaluated directly against alternative tools in terms of implementation efficiency and workflow scope.

5 Availability

PALM-mGWAS is distributed as the PALMmGWAS R package together with shell workflow wrappers, example input data, and documentation. The implementation supports local execution through `pixi` and containerized deployment in high-performance computing environments.

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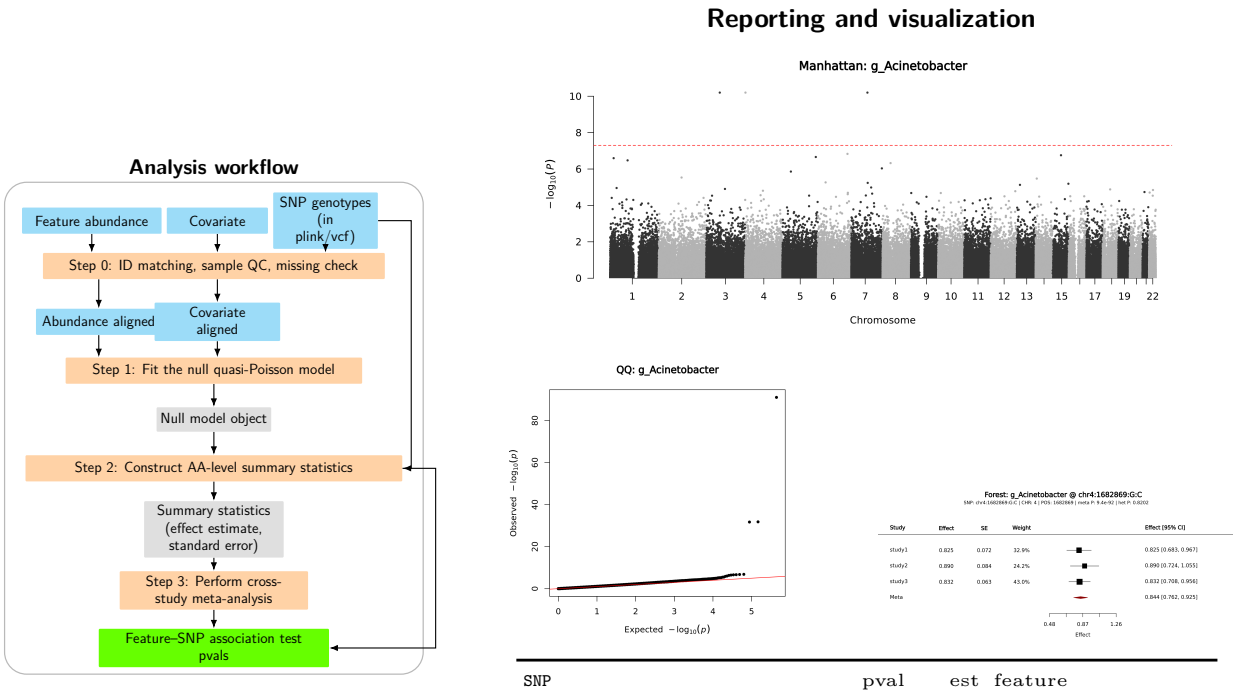
Conflict of Interest

None declared.

References

Movassagh, M. R. *et al.* (2023) *mbQTL*: an R/Bioconductor package for microbial quantitative



SNP	pval	est	feature
chr4:1682869:G:C	9.38×10^{-92}	0.844	g__Acinetobacter
chr7:81941667:G:C	1.57×10^{-32}	0.529	g__Acinetobacter
chr3:68821275:C:T	1.95×10^{-32}	0.646	g__Acinetobacter
chr15:54854424:C:T	1.54×10^{-8}	0.239	f__Bifidobacteriaceae

Figure 1: End-to-end analysis and reporting workflow of PALM-mGWAS. Left: the PALM-based analysis pipeline is drawn directly in LaTeX to summarize preprocessing, null-model fitting, AA-level summary-statistic construction, and cross-study meta-analysis. Right: representative reporting outputs generated from the current example dataset, including a feature-level Manhattan plot, the corresponding QQ plot, a selected-SNP forest plot, and the filtered combined-hit summary table.

trait loci (QTL) estimation. *Bioinformatics*.